Day 144: Prompt Engineering for Generative Models: Real Experimentation

MD Mutasim Billah | 52-Week Data Science to ML/AI Roadmap | Week 21, Day 5 | Complete study guide

Watch: Classifier-Free Guidance Diffusion Models for Image Synthesis (AI Focus) | Mastering Negative Prompts in Stable Diffusion (Making AI Magic)

Script: `day144_prompt_engineering.py` · every number in this guide is copied by program from `day144_run_output.txt`, the real output of that script.

Day 143's text encoder now steers Day 141's diffusion model. The network gets a caption's 64-number embedding as an extra input and is trained on captioned digits, with the caption dropped 15% of the time so that the same network also learns to draw with no caption. At sampling time, classifier-free guidance pushes the captioned guess away from the uncaptioned one. With no push the prompt is ignored (the right digit 0.107 of the time, chance); with guidance 2 every sample is the right digit (1.000) at the best FID (25.06); stronger guidance makes the samples worse and less varied. Prompts were then varied one change at a time: paraphrases, unknown words, attributes, a combination never seen in training, negative prompts and contradictions. Each change was measured on 200 drawings.

0. Where today fits

DEFINITION — start here, no prior knowledge assumed

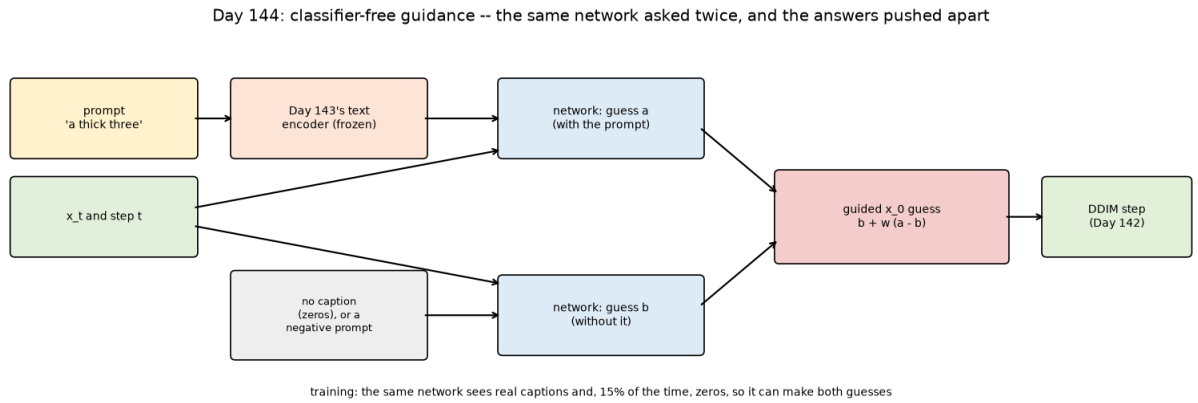
A **prompt** is the text a user gives a text-to-image model. **Prompt engineering** is choosing that text, and the settings around it, to get the image wanted. Its effects can be measured: draw many images from each prompt and count what comes out.

Classifier-free guidance (Ho and Salimans, 2022) is how most text-to-image diffusion models make the prompt count. One network is trained both with captions and with an empty caption. At each sampling step it is asked twice, with and without the prompt, and the guess used is the uncaptioned one plus w times the difference. The **guidance scale** w sets how hard the prompt pushes; a **negative prompt** replaces the empty caption, so the push is away from it.

REAL-WORLD SCENARIO

A designer asks an image generator for 'a red bicycle on a beach'. The result ignores the beach, so they turn up the guidance scale; the bicycle and beach appear, but the colours become harsh and every image looks alike. They add 'blurry' as a negative prompt and the images sharpen. Every one of those moves is measured below, on digits.

SYSTEM DIAGRAM



Classifier-free guidance: the same network is asked with the prompt and without it (or with a negative prompt), and the guided guess goes past the prompted one.

Today's part for the week's build

`day144_exercise.py` is today's parts (the captioned guess, the training loop with dropped captions, guided DDIM sampling with an optional negative prompt, the two measurements, and naming digits), checked by hand: w = 1 with the empty caption spelled out reproduces the one-call shortcut exactly, a made-up stroke measures as expected, and a briefly trained model already draws thicker digits for 'zero' than for 'one'.

ACTUAL OUTPUT

```
w = 1 with 'no caption' passed explicitly against the one-call shortcut: largest difference
    0.00e+00
a made-up stroke: ink 0.0357 (28 / 784 = 0.0357), lean 0.335
1,500 steps of training: loss 0.1939 -> 0.1499
guided samples, w = 2: 'zero' ink 0.2040, 'one' ink 0.0654 (real training digits: 0.1734 and
    0.0760)
```

Data: Day 126's 70,000 MNIST digits (training images 0–59,999; pixels scaled to [–1, 1] for the diffusion model). Captions are Day 143's, written from each digit's label, thickness and lean; Day 143's encoders are retrained here with the same held-out pair (a seven is never called thick). All samples use Day 142's DDIM with 100 steps and no fresh noise; Day 138's classifier names the digits and its scorer gives FID, precision and recall.

How to read each section

Step	What it gives you
DEFINITION	What the idea is, in plain words. Start here.
REAL-WORLD SCENARIO	An everyday situation where the same idea shows up.
MATHEMATICS	The formula, when the section introduces one.
SPECIFICATION	The exact setup of the experiment.
IMPLEMENTATION	The real code from the script.
ACTUAL OUTPUT	What the script really printed, copied unchanged.
What it means	How to read the numbers, and what they do and do not show.
Why choose this? / Why NOT this?	When the idea is the right tool, and what it costs.

1. By hand: guidance in the x_0 guess is guidance in the noise guess

DEFINITION — start here, no prior knowledge assumed

Papers write guidance on the noise guess ε; this model's output is a guess of the clean digit x_0 (Day 141). Because the noise implied by an x_0 guess is a straight-line function of it, mixing two x_0 guesses and then converting gives exactly the same answer as converting both and then mixing. So guidance can be applied to either.

REAL-WORLD SCENARIO

Converting two prices to another currency and then taking their difference gives the same answer as taking the difference first and converting it: the conversion is a fixed rate.

MATHEMATICS

$$\text{guided guess} = b + w\,(a - b), \quad a = f(x_t, t, \text{prompt}), \quad b = f(x_t, t, \text{no caption})$$

$$\varepsilon(x_0) = (x_t - \sqrt{A_t}\,x_0) / \sqrt{(1 - A_t)} \Rightarrow \varepsilon(b + w(a - b)) = \varepsilon(b) + w\,(\varepsilon(a) - \varepsilon(b))$$

$w = 0$: no caption; $w = 1$: plain conditioning; $w > 1$: past the prompt; negative prompt: b uses its embedding

IMPLEMENTATION

```
def cfg_sample(net, cpos, n, w, cneg=None, steps=SAMPLE_STEPS, seed=1):
    cp = np.repeat(cpos[None], n, 0)
    cn = np.zeros_like(cp) if cneg is None else np.repeat(cneg[None], n, 0)
    def x0_fn(x, t):
        tt = np.full(len(x), t); a = guess_c(net, x, tt, cp)[1]
        if w == 1.0 and cneg is None:
            return a
        b = guess_c(net, x, tt, cn)[1]
        return b + w * (a - b)
    return d141.to_pixels(d142.ddim_sample(x0_fn, n, np.random.RandomState(seed), steps, 0.0))
```

ACTUAL OUTPUT

```
eps(x_0) = (x_t - sqrt(abar_t) x_0) / sqrt(1 - abar_t) is a straight-line function of x_0, so mixing guesses commutes with
    it:
  w = 0.0: largest difference between eps(guided x_0) and guided eps: 0.00e+00
  w = 1.0: largest difference between eps(guided x_0) and guided eps: 4.44e-16
  w = 3.0: largest difference between eps(guided x_0) and guided eps: 1.78e-15
w = 0 returns the no-caption guess exactly, w = 1 the captioned guess exactly; a negative prompt simply takes the no-caption
    guess's place
Actual output from this lesson's run.
```

What it means

The two orders agree to rounding error (0.00e+00 at w = 0, 1.78e-15 at w = 3). The code therefore mixes x_0 guesses directly, and Day 142's DDIM turns the result into a step. With w = 1 and no negative prompt the second network call is skipped: the guess is just the captioned one.

WHY CHOOSE THIS?

One network, no separate classifier, and a single number that sets how strongly the prompt is followed.

WHY NOT THIS? (TRADEOFFS)

Two network calls per step whenever $w \neq 1$: twice the sampling cost.

2. Training the text-conditioned model

DEFINITION — start here, no prior knowledge assumed

The network is Day 141's (two hidden layers of 512, output read as x_0, output layer started at zero) with 64 more inputs: the caption's embedding. Each training digit gets a fresh random caption, embedded by Day 143's text encoder, which is trained first and then frozen. 15% of the time the embedding is replaced by zeros, standing for 'no caption'.

REAL-WORLD SCENARIO

Teaching an artist from a pile of sketches where most have a note saying what they show and some have none: the artist learns both to draw what a note asks and to draw something plausible when there is no note.

SPECIFICATION

Network	x_t (784) + time embedding (32) + caption (64) $\rightarrow$ 512 $\rightarrow$ 512 $\rightarrow$ 784, ReLU
Captions	Day 143's: digit word, sometimes thickness and lean, a filler; thick sevens never called thick
Training	Adam lr 0.001, batches of 128, 20,000 steps; 15% of captions dropped; averaged copy of the weights (decay 0.999), used for all sampling

IMPLEMENTATION

```
def train_cond(X2, y, thick, lean, txt, steps=STEPS, drop=DROP, hold=HOLD, seed=0, batch=128,
    lr=1e-3, every=2000):
    net = d126.init_net(SIZES, seed); net[-1] = (net[-1][0] * 0, net[-1][1])
    opt = d126.Adam(net, lr); ema = [(W.copy(), b.copy()) for W, b in net]
    rng = np.random.RandomState(seed); log = []; run = 0.0
    for s in range(1, steps + 1):
        i = rng.randint(0, len(X2), batch); x0 = X2[i].astype(np.float64); t = rng.randint(1,
            len(ABAR), batch)
        e = rng.standard_normal(x0.shape); xt = np.sqrt(ABAR[t])[:, None] * x0 + np.sqrt(1 -
            ABAR[t])[:, None] * e
        c = d143.embed_texts(txt, [d143.caption(y[k], thick[k], lean[k], rng, hold) for k in i])
        c[rng.rand(batch) < drop] = 0.0
        A, x0h = guess_c(net, xt, t, c); miss = x0h - x0
        grads, _ = d126.backward(net, A, ACTS, (miss * 2 / miss.size).astype(np.float32))
        opt.step(net, grads); d142.ema_update(ema, net); run += (miss ** 2).mean()
        if s % every == 0:
            log.append((s, run / every)); run = 0.0
    return net, ema, log
```

ACTUAL OUTPUT

```
Day 143's encoders, trained with thick sevens never called thick: 21 s
diffusion model: Day 141's network with 64 more inputs (880 -> 512 -> 512 -> 784), output read as x_0, output layer started
    at zero;
each training digit gets a fresh random caption (Day 143's, the same held-out pair), embedded by the text encoder, which is
    frozen;
15% of captions replaced by zeros ('no caption'); Adam lr 0.001, batches of 128, 20,000 steps, averaged copy decay 0.999
    step     loss
   2,000   0.1620
   4,000   0.1411
   6,000   0.1375
   8,000   0.1358
  10,000   0.1350
  12,000   0.1341
  14,000   0.1334
  16,000   0.1331
  18,000   0.1327
  20,000   0.1320
(559 s, including the text encoder embedding every caption)
Actual output from this lesson's run.
```

What it means

The loss fell from 0.1620 to 0.1320 over the 20,000 steps, most of it in the first 6,000. Training took 559 s, including embedding a caption for every training digit. As on Day 141, the loss alone says nothing about whether the prompts work; sections 3–7 measure that directly.

WHY CHOOSE THIS?

A captioned model is the uncaptioned one with 64 more inputs; dropping captions costs nothing extra and gives guidance for free.

WHY NOT THIS? (TRADEOFFS)

Every caption the model will understand must appear, in some form, in the training captions.

3. The guidance scale: following the prompt against quality and variety

DEFINITION — start here, no prior knowledge assumed

The ten prompts 'zero' to 'nine' each draw 200 digits at six guidance scales. Day 138's classifier names each drawing; 'right digit' is the share named as the prompted digit. FID, precision and recall compare all 2,000 with real digits; the grey share (Day 142) measures blur and ink the average pixel.

REAL-WORLD SCENARIO

Turning up the volume on a radio with a weak station: at first the voice gets clearer; turn it too far and it distorts.

ACTUAL OUTPUT

```
prompts 'zero' ... 'nine', 200 samples each (2,000 in all), DDIM 100 steps; 'right digit': the share Day 138's classifier
names as the prompted digit (0.10 is chance); FID, precision and recall against real training digits; ink: the average pixel
    w  right digit     IS      FID  precision  recall  grey share    ink   time
  0.0        0.107   4.64    72.99      0.819   0.476      0.1251  0.124    12s
  1.0        0.948   8.52    25.47      0.924   0.527      0.1097  0.128     8s
  2.0        1.000   9.90    25.06      0.930   0.464      0.0848  0.150    12s
  3.0        1.000   9.98    70.09      0.856   0.341      0.0678  0.166    12s
  5.0        1.000   9.98   142.70      0.694   0.190      0.0463  0.183    12s
  8.0        1.000   9.96   210.96      0.571   0.345      0.0275  0.189    12s
 real                                                     0.0500  0.132   (2,000 real test digits)
Actual output from this lesson's run.
```

What it means

Without guidance (w = 0) the prompt is ignored: right digit 0.107, chance. Plain conditioning (w = 1) already follows it 0.948 of the time, and w = 2 gave both every digit right (1.000) and the best FID (25.06). Beyond that, quality fell fast: FID 70.09 at w = 3, 142.70 at 5 and 210.96 at 8, with recall falling from 0.527 to 0.190 at w = 5.

Two quieter changes came with guidance. The digits got thicker (ink 0.124 at w = 0, 0.189 at w = 8, against 0.132 for real digits) and sharper (grey share 0.1251 down to 0.0275, against 0.0500). So guidance does more than make a model follow the prompt. A likely reason, not tested here: it pushes every image towards what most distinguishes the prompt, and here that included heavier strokes. Day 145 measures this kind of exaggeration on purpose.

WHY CHOOSE THIS?

One number trades how closely the prompt is followed against quality and variety, at sampling time, with no retraining.

WHY NOT THIS? (TRADEOFFS)

The best value (here 2) has to be found by measuring; past it, images degrade and lose variety.

4. Paraphrases, and words the model does not know

DEFINITION — start here, no prior knowledge assumed

The same request worded nine ways, from 'seven' to 'a photo of the number seven', plus a prompt with irrelevant words ('seven purple elephant') and two the model cannot read: '7' and 'sept'. All at w = 2, 200 drawings each, from the same starting noise.

REAL-WORLD SCENARIO

Ordering coffee in a café abroad: 'a coffee', 'coffee please' and 'one coffee' all work; asking in a language the barista does not speak gets you whatever they make by default.

ACTUAL OUTPUT

```
200 samples per prompt from the same starting noise (seed 1), guidance w = 2.0; 'right': the share named as the intended
    digit;
ink and lean: Day 143's measurements, averaged; classified as: the two digits Day 138's classifier names most often
                        prompt  negative prompt   right     ink    lean    grey   classified as
                      'seven'                -   1.000  0.1497  0.0160  0.0699   seven 200, zero 0
                    'a seven'                -   1.000  0.1500  0.0159  0.0712   seven 200, zero 0
             'the digit seven'                -   1.000  0.1490  0.0154  0.0690   seven 200, zero 0
           'handwritten seven'                -   1.000  0.1506  0.0133  0.0698   seven 200, zero 0
             'seven written a'                -   1.000  0.1530  0.0249  0.0699   seven 200, zero 0
 'a photo of the number seven'                -   1.000  0.1334  0.0063  0.0852   seven 200, zero 0
       'seven purple elephant'                -   1.000  0.1497  0.0160  0.0699   seven 200, zero 0
                           '7'                -   0.000  0.1083  0.1446  0.1113   eight 84, five 24
                        'sept'                -   0.000  0.1083  0.1446  0.1113   eight 84, five 24
'7' against an empty caption: cosine 1.000000; an empty caption against 'no caption' (zeros, used in training): different
    (largest entry 0.274)
real training sevens: ink 0.1145, lean 0.0241
Actual output from this lesson's run.
```

What it means

Every wording that contains 'seven' gave 200 sevens out of 200. Filler words changed the drawings only slightly (ink 0.1497 to 0.1530); 'a photo of the number seven', with three words the encoder does not know, gave thinner sevens (0.1334). 'seven purple elephant' gave exactly the same drawings as 'seven': unknown words are dropped.

'7' and 'sept' gave no sevens at all (0.000): mostly eights (eight 84). An unknown prompt is an empty caption ('7' against an empty caption: cosine 1.000000), and an empty caption is not 'no caption': the text encoder turns it into a fixed vector of its own, which the model never saw in training and reads as something eight-like. A model can fail silently on a prompt it cannot read, by drawing something confident and wrong.

WHY CHOOSE THIS?

Paraphrases that use known words are interchangeable, so wording can be chosen for the reader, not the model.

WHY NOT THIS? (TRADEOFFS)

Words outside the training captions are not ignored safely: here they turned a request for a seven into eights.

5. Attributes, and a combination never seen in training

DEFINITION — start here, no prior knowledge assumed

Prompts that add thickness or lean to a digit, each measured with Day 143's ink and lean, against the plain digit and real training digits. 'a thick seven' is the held-out pair: neither the text encoder's captions nor the diffusion model's ever called a seven thick.

REAL-WORLD SCENARIO

Asking a sign-writer for 'the same letter, but bold' or 'in italics': the request changes one property and keeps the rest.

ACTUAL OUTPUT

```
200 samples per prompt from the same starting noise (seed 1), guidance w = 2.0; 'right': the share named as the intended
    digit;
ink and lean: Day 143's measurements, averaged; classified as: the two digits Day 138's classifier names most often
                       prompt  negative prompt   right     ink    lean    grey   classified as
                      'three'                -   1.000  0.1690  0.0808  0.0864   three 200, zero 0
               'a thick three'                -   1.000  0.2302  0.0185  0.0824   three 200, zero 0
                'a thin three'                -   1.000  0.1032  0.0383  0.0841   three 200, zero 0
                       'four'                -   1.000  0.1379  0.0591  0.0802   four 200, zero 0
          'four leaning right'                -   1.000  0.1468  0.4722  0.0760   four 200, zero 0
           'four leaning left'                -   0.985  0.1401 -0.2718  0.0755   four 197, nine 3
                'four upright'                -   1.000  0.1404  0.0949  0.0793   four 200, zero 0
                       'seven'                -   1.000  0.1497  0.0160  0.0699   seven 200, zero 0
               'a thick seven'                -   0.990  0.2111  0.0012  0.0832   seven 198, four 1
                'a thin seven'                -   1.000  0.0906  0.0109  0.0581   seven 200, zero 0
real training threes: ink 0.1415, lean 0.0555
real training fours: ink 0.1214, lean 0.0576
real training sevens: ink 0.1145, lean 0.0241
'a thick seven': neither model's training captions ever put 'thick' with 'seven'
Actual output from this lesson's run.
```

What it means

Attributes steered the drawings strongly. 'a thick three' drew ink 0.2302 and 'a thin three' 0.1032, against 0.1690 for 'three' and 0.1415 for real threes; 'four leaning right' leaned 0.4722 and 'four leaning left' -0.2718, against 0.0591 for 'four'.

The held-out 'a thick seven' worked: ink 0.2111 against 0.1497 for 'seven', with seven 198 of 200 still sevens. As on Day 143, the bag-of-words encoder adds the meaning of 'thick' to that of 'seven', and the diffusion model had learned to draw 'thick' from other digits.

WHY CHOOSE THIS?

Properties can be asked for one at a time and combined, including combinations never seen together.

WHY NOT THIS? (TRADEOFFS)

Only properties named in the training captions can be asked for; nothing here tests a property the captions never mention.

6. Negative prompts

DEFINITION — start here, no prior knowledge assumed

A negative prompt takes the place of the empty caption in guidance, so each step is pushed away from it as well as towards the prompt. Here: 'three' with 'thick' or 'thin' as the negative, 'four' with a lean, and two digits as negatives.

REAL-WORLD SCENARIO

Telling a hairdresser 'not too short': it does not say what to do, only what to move away from, and they cut a little longer than usual.

ACTUAL OUTPUT

```
6. NEGATIVE PROMPTS -- guide away from a second prompt instead of from 'no caption'
======================================================================
200 samples per prompt from the same starting noise (seed 1), guidance w = 2.0; 'right': the share named as the intended
    digit;
ink and lean: Day 143's measurements, averaged; classified as: the two digits Day 138's classifier names most often
                        prompt  negative prompt   right     ink    lean    grey   classified as
                      'three'               -   1.000  0.1690  0.0808  0.0864   three 200, zero 0
                      'three'          'thick'   0.990  0.0810  0.0915  0.0659   three 198, five 2
                      'three'           'thin'   0.975  0.2663  0.0007  0.0498   three 195, eight 3
                       'four'  'leaning right'   0.970  0.1563 -0.1674  0.0614   four 194, nine 6
                       'four'   'leaning left'   0.955  0.1634  0.4064  0.0584   four 191, eight 5
                      'three'          'eight'   1.000  0.1758  0.0476  0.0772   three 200, zero 0
                        'one'          'seven'   1.000  0.0878  0.2335  0.0401   one 200, zero 0
Actual output from this lesson's run.
```

What it means

Negative prompts worked, and strongly. 'three' with negative 'thick' drew ink 0.0810, thinner than 'a thin three' (0.1032); with negative 'thin', 0.2663, thicker than 'a thick three' (0.2302). 'four' with negative 'leaning right' leaned left (-0.1674). A likely reason it is stronger, not tested here: the two guesses differ only in the attribute, so the whole push goes into it.

The cost is the digit itself: the right digit fell to 0.975 with negative 'thin' and 0.955 with negative 'leaning left' (some came out as eights and nines). Without the empty caption in the guidance, nothing pushes away from 'any digit at all'.

WHY CHOOSE THIS?

A direct way to remove an unwanted property without having to name the wanted one.

WHY NOT THIS? (TRADEOFFS)

Strong, hard to judge in advance, and here it cost a few digits their identity.

7. Contradictory prompts

DEFINITION — start here, no prior knowledge assumed

Prompts that ask for two things at once: two digits, or two opposite attributes.

REAL-WORLD SCENARIO

Asking a taxi driver for 'the airport and the station': they will take you to one of them, not to a point halfway between.

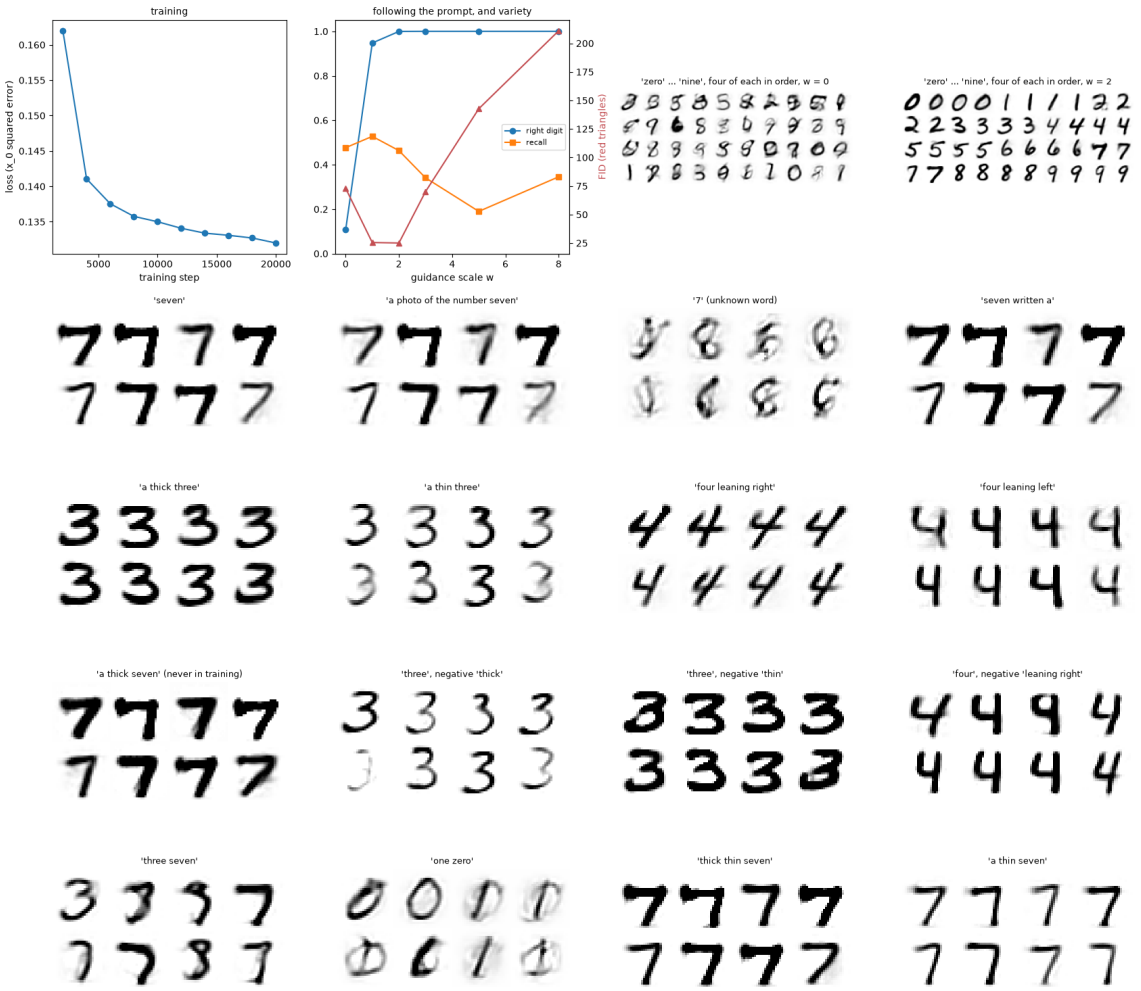
ACTUAL OUTPUT

```
7. CONTRADICTORY PROMPTS
======================================================================
200 samples per prompt from the same starting noise (seed 1), guidance w = 2.0; 'right': the share named as the intended
    digit;
ink and lean: Day 143's measurements, averaged; classified as: the two digits Day 138's classifier names most often
                        prompt  negative prompt   right     ink    lean    grey   classified as
                 'three seven'               -       -  0.1432 -0.0423  0.0885   three 112, seven 86
                 'three eight'               -       -  0.1515  0.1561  0.1073   eight 169, three 29
                    'one zero'               -       -  0.1206  0.2852  0.1065   zero 88, one 86
             'thick thin seven'               -       -  0.1558 -0.0247  0.0519   seven 200, zero 0
     'leaning left right four'               -       -  0.1310  0.1393  0.0806   four 200, zero 0
Actual output from this lesson's run.
```

What it means

Two digits gave one or the other, not a blend: 'three seven' gave three 112, seven 86; 'one zero' gave zero 88, one 86. 'three eight' mostly gave eights (eight 169). A likely reason, not tested here: each sample commits early in sampling to one of the two digits; the figure shows a few that look like neither.

Opposite attributes cancelled: 'thick thin seven' drew ink 0.1558, close to plain 'seven' (0.1497). In a bag of words the two words are averaged; a likely reason, not tested here, is that their opposite directions largely cancel.



Top: training loss; right digit, recall and FID against the guidance scale; four drawings for each prompt 'zero' ... 'nine' in order, at w = 0 and w = 2. Below: 8 drawings each for sections 4–7's prompts, from the same starting noise.

WHY CHOOSE THIS?

Shows what the model does when a request cannot be satisfied: useful to know before users find out.

WHY NOT THIS? (TRADEOFFS)

With a bag-of-words encoder, 'and', 'not' and word order mean nothing; a real text encoder would behave differently.

Interview preparation — technical questions

Q: Write classifier-free guidance.

A: Guided guess = b + w (a – b), a with the prompt, b with the empty caption; here w = 2 gave right digit 1.000 and FID 25.06.

Q: How is a model trained for classifier-free guidance?

A: Drop the caption for a share of training examples (15% here), replacing its embedding with zeros, so one network learns both.

Q: What does a negative prompt do mechanically?

A: It replaces the empty caption in b; 'three' with negative 'thick' drew ink 0.0810, thinner than 'a thin three' (0.1032).

Q: How would you choose the guidance scale?

A: Sweep it and measure: here prompt-following saturated at w = 2 while FID was best there and rose to 210.96 at w = 8.

Q: How do you test a prompt change fairly?

A: Same starting noise, many samples, and a measurement of the property: here 200 drawings per prompt from seed 1.

Interview preparation — theoretical questions

Q: Why does high guidance reduce variety?

A: Measured: recall fell from 0.527 to 0.190 at w = 5. A likely reason, not tested here: every sample is pushed towards what most separates the prompt from no prompt, so samples converge.

Q: Why can x_0 guesses be guided like noise guesses?

A: The noise implied by an x_0 guess is a straight-line function of it, so mixing commutes with converting (difference 1.78e-15).

Q: What happens with words a model cannot read?

A: Here they became an empty caption, which is not 'no caption': '7' drew eights, not sevens.

Q: Why did 'three seven' give threes and sevens, not a blend?

A: A likely reason, not tested here: each sample commits early in sampling to one of the two digits it is pushed towards.

Q: What did guidance do besides follow the prompt?

A: Measured: it thickened the digits (ink 0.124 to 0.189, against 0.132 for real digits), past what the data shows.

Key takeaway and what's next

> **KEY TAKEAWAY**
>
> A prompt steers a diffusion model through one extra input and one formula. Guidance at w = 2 made every drawing follow its prompt with the best quality; more guidance made drawings worse, less varied and heavier. Paraphrases with known words were interchangeable; unknown words silently produced the wrong digit. Attributes, combinations never seen together and negative prompts all worked, and were measured; contradictions picked a side. Prompt engineering is measurable, and should be measured.

- Guidance scale: right digit 0.107 (w = 0), 0.948 (1), 1.000 (2); FID 25.06 at 2, 210.96 at 8.
- '7' and 'sept': no sevens; mostly eights.
- 'a thick three' ink 0.2302, 'a thin three' 0.1032; 'a thick seven' (never in training) 0.2111.
- 'three' with negative 'thick': ink 0.0810; with negative 'thin': 0.2663.
- 'three seven': three 112, seven 86; 'thick thin seven': ink 0.1558, close to 'seven' (0.1497).

Suggested watch order, if starting from zero

- 1. Classifier-Free Guidance Diffusion Models for Image Synthesis (AI Focus) — how guidance works in real text-to-image models.
- 2. Mastering Negative Prompts in Stable Diffusion (Making AI Magic) — negative prompts in practice, in Stable Diffusion.

Day 145 preview

- **Ethical considerations in generative AI**: a deliberately skewed training set and what guidance does to the skew, a detector of generated images and how it fails, memorisation with tiny training sets, and four documented real cases.